How Fragile Is Your Vocabulary? A Controlled Audit of Inference-Vocabulary Robustness in Training-free Open-Vocabulary Semantic Segmentation

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Abstract

Training-free open-vocabulary semantic segmentation (OVSS) is evaluated as if the inference vocabulary were a fixed, clean, given input, yet the promise of the task is precisely that users supply arbitrary class names at test time. We present a controlled audit of inference-vocabulary robustness. We freeze a single inference protocol and perturb only the vocabulary along three orthogonal axes—synonym substitution, granularity shift, and distractor injection—across three representative training-free methods (MaskCLIP, SCLIP, ClearCLIP; key cells replicated on NA-CLIP) and four benchmarks (VOC-21, COCO-171, ADE-150, PC-459); the core audit matrix comprises about 150 evaluation runs, and the full study archives over 600. The audit yields five findings. (1) Undocumented class-name engineering in official vocabularies is worth 11.8–20.6 mIoU on VOC-21 across the three main-matrix methods (up to 22.1 on the extended seven-method leaderboard), comparable to or exceeding the gaps between methods. (2) WordNet-filtered synonym substitution costs up to 6 mIoU on the three main-matrix methods (up to 7 in the replication and backbone checks), with a 3.7 mIoU spread across random substitution seeds, and one-level hypernym substitution halves VOC-21 and nearly zeroes ADE-150 performance. (3) Injecting distractor classes *raises* mIoU over ground-truth classes by up to 10 points while lowering an object-crop classification control by 18.5 points; the gain reverses into a drop once the engineered background vocabulary is used, isolating background under-modelling as the mechanism and showing that dense vocabulary sensitivity cannot be read off classification behaviour. (4) Pre-registered text-side repairs (mean-centering, vocabulary-conditioned and global ZCA whitening) improve embedding-space separability metrics monotonically yet reduce mIoU in 8 of 9 method×dataset cells on mid-to-large vocabularies, showing that text embedding geometry is not a reliable proxy for segmentation quality. (5) A constructive attempt to automate vocabulary engineering recovers most of its macro-mIoU value (up to +18.8 on plain vocabularies) with synthesised background negatives, but every pre-registered label-free mechanism for its per-class protective component fails, indicating engineered vocabularies encode supervision that text-side machinery cannot reconstruct. We release the perturbation protocol, vocabularies, and evaluation harness to make vocabulary robustness a reportable, comparable quantity.

1 Introduction

Open-vocabulary semantic segmentation promises segmentation for class sets specified freely at inference time. A rapidly growing family of *training-free* methods [1–5] adapts CLIP [8] to dense prediction without any segmentation training, and is evaluated by zero-shot mIoU on standard

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benchmarks. In every such evaluation, the inference vocabulary—the list of class names presented to the text encoder—is treated as a fixed part of the benchmark. The open half of "open-vocabulary" is never itself tested.

This gap matters because the vocabulary is the only input the user controls. If performance and method rankings change materially under semantically equivalent or mildly perturbed vocabularies, then reported numbers measure a particular prompt engineering artifact rather than open-vocabulary capability. Anecdotal evidence already exists: public implementations ship hand-tuned class-name files (e.g. expanding the VOC "background" class into 26 scene sub-classes [2]), and prompt augmentation methods [9, 10] show that name phrasing shifts classification accuracy. What is missing is a controlled, dense-prediction-specific audit that isolates the vocabulary as the sole experimental variable.

We provide such an audit. We freeze one inference protocol—backbone, templates, resolution, sliding window, logit pooling—and vary only the vocabulary along three orthogonal axes: (i) *synonym substitution* with WordNet+CLIP-similarity filtering at 25/50/100% replacement rates and three seeds; (ii) *granularity shift* via a frozen first-sense WordNet hypernym rule; (iii) *distractor injection* of 50–200 out-of-dataset class names stratified by text similarity. We evaluate three representative training-free methods on four benchmarks spanning 21 to 459 classes: VOC-21 [34], COCO-171 [36, 37], ADE-150 [38], and PC-459 [35]. Alongside the audit we pre-registered two text-side repair hypotheses with kill criteria: vocabulary-conditioned whitening of class embeddings, and its comparison against global-statistics whitening. KC1 failed outright; KC2's literal criterion is met only on COCO-171, where both whitening variants remain well below the unwhitened baseline, so no usable repair emerges. We report these failures as findings.

Our contributions are:

1. **A frozen three-axis vocabulary perturbation protocol** for OVSS, with released perturbed vocabularies, exclusion rules for non-semantic classes, and a uniform inference protocol that removes per-method prompt engineering.

2. **An audit of three training-free methods on four benchmarks** showing that (a) undocumented name engineering is worth 11.8–20.6 mIoU (up to 22.1 on the extended leaderboard of Table 6), comparable to or larger than method-to-method gaps; (b) synonym-seed spread reaches 3.7 mIoU, larger than many reported improvements; (c) the relative ordering of methods is not stable across vocabulary regimes (e.g. ClearCLIP leads SCLIP by 2.5 mIoU on PC-459 while trailing by 1.8 under the official VOC-21 vocabulary).

3. **Dense-specific evidence**: under distractor injection, an object-crop classification control drops by 18.5 points while segmentation mIoU *rises* by up to 10 points—and the gain reverses under the engineered background vocabulary—exposing a background-absorption pathology of the mIoU protocol; vocabulary robustness conclusions cannot be imported from classification.

4. **A negative result on text-side repair**: mean-centering and ZCA whitening (vocabulary-conditioned or global; verdict pre-registered at shrinkage 0.5 and confirmed by a post-verdict shrinkage sweep) improve embedding margin and effective rank monotonically yet decrease mIoU in 8 of 9 mid-to-large-vocabulary cells—embedding-geometry diagnostics are not a proxy for segmentation quality.

5. **A constructive decomposition of vocabulary engineering** (§4.5): its macro-mIoU value is largely an automatable background-sink effect (synthesised negatives recover +16.1 macro on VOC-21 plain vocabularies without harming official ones), while its per-class protective

component resisted every pre-registered label-free mechanism (three signal families in the main text; seven entries in the appendix index), isolating exactly where label-derived knowledge enters current benchmarks.

2 Related Work

Training-free OVSS. Open-vocabulary segmentation was first pursued with training: language-aligned decoders and mask classifiers fine-tuned on segmentation or image-text data [52–57], or learned from image-text pairs alone [58, 59]. The training-free line instead re-reads a frozen CLIP [8]: MaskCLIP [1] extracts dense predictions from CLIP by converting the final attention pooling into a value-only pathway; SCLIP [2] uses correlative self-attention ($\mathrm{softmax}(qq^\top)+\mathrm{softmax}(kk^\top)$); ClearCLIP [3] additionally drops the residual path and FFN; NACLIP [5] enforces neighbourhood-aware attention via a Gaussian spatial prior; ProxyCLIP [4] imports affinities from visual foundation models; GEM [27] and CLIP-DINOiser [28] are further representatives. All are evaluated on fixed official vocabularies; none reports vocabulary sensitivity. We audit MaskCLIP, SCLIP, and ClearCLIP as representative final-block variants under one protocol, and scope our conclusions to this family (§6).

Text-side sensitivity and name quality. On the classification side, prompt template ensembles [8], learned soft prompts [51], LLM-generated descriptions [10], random descriptors [9], WordNet-hierarchy augmentation [24], concept-frequency analyses [25], and synonym-space aggregation [13] all show that name phrasing shifts zero-shot accuracy; vocabulary-free classification [26] questions the given-vocabulary assumption itself. For segmentation, RENOVATE [12] shows benchmark class names are often misaligned with masks and that *renaming* them changes results substantially, and FLOSS [14] shows per-class template selection alone improves a frozen segmenter; SynCLIP [21] confirms synonym-induced grounding inconsistency in dense perception and repairs it by synonym-coherent *re-pretraining* — adapting the model to the names, whereas we freeze the model and perturb the names, so our protocol can audit SynCLIP-class models directly. A data-centric retrieval family (ReME [20]) replaces attention surgery with reference-set retrieval; its released repository contains no method code at audit time, so it is not among our anchors, but its name-mediated retrieval interface makes it a natural future audit subject. A training-free class-purification plugin (FreeCP [19]) manipulates the query set itself, making it a natural interlocutor for our distractor axis; however, its released code, in its official configuration (the LLM dictionaries are consumed unconditionally inside its purification step, with no non-LLM mode), requires precomputed per-name LLM (Vicuna-13B) description dictionaries keyed to the official engineered class names, so at audit time a plain, synonym, or distractor vocabulary cannot even be expressed without regenerating those assets — vocabulary engineering one level deeper than the class-name files we perturb — and it therefore cannot be audited under our only-change-the-name-files protocol. We differ on both axes of intent: these works *improve* names or prompts toward a better fixed evaluation, whereas we hold labels fixed and *measure the response* of methods to controlled, user-plausible vocabulary perturbations—including axes (distractor injection, granularity, seed variance of synonym choice) that name-improvement work does not quantify—plus pre-registered tests showing that generic whitening-style repairs do not transfer to dense prediction.

Background modelling in OVSS. That an explicit background model matters is folklore in the method literature: CaR [6] adds hand-written background queries, OVDiff [7] derives negative visual prototypes from a diffusion model, and SCLIP's official VOC vocabulary expands "background" into

26 scene sub-classes [2]. Our constructive experiment (§4.5) differs in intent from these methods: rather than proposing another background mechanism, it uses pre-registered text-side background synthesis—with random-word controls and per-class safety criteria—to decompose *how much* of the engineered-vocabulary value such mechanisms can recover without labels, and which component cannot.

Embedding geometry. CLIP text embeddings concentrate in a narrow cone; for sentence embeddings, whitening improves STS performance and speeds up retrieval via dimensionality reduction [11]. We confirm the cone (mean pairwise cosine 0.77–0.82 across vocabularies) and show that removing it helps linear separability metrics but not segmentation, a dissociation previously unreported.

Evaluation audits and metrics. MESS [23] audits zero-shot segmentation across *domains*; our audit varies the complementary input, the vocabulary. Zhou et al. [18] show pixel-wise IoU mishandles synonym/near-synonym predictions and propose Open mIoU, and Huang et al. [15] study vocabulary expansion with semantically ambiguous words and propose a mask-wise protocol; our distractor results (§4.3) explain when expansion raises rather than lowers the reported number—the direction is governed by background modelling and metric convention—and support adopting such open-set-aware metrics. A concurrent "systematic audit" of VLM robustness to natural semantic variation [22] targets classification/retrieval under typographic and semantic variants, not dense segmentation or inference-vocabulary perturbation. In remote sensing, the concurrent OVEarth-Bench [33] scores open-vocabulary Earth-observation models on category breadth and query diversity within a fixed benchmark vocabulary (negatives are absent classes to be rejected); it does not perturb the names of a fixed target class, which is the axis audited here. Our motivation parallels the classification test-set audits that asked whether reported gains survive a change of the nominally-fixed evaluation input [60, 61]: there the images, here the names. To our knowledge no prior work performs a controlled multi-axis inference-vocabulary perturbation audit across training-free OVSS methods; the closest works study name quality [12] or template choice [14] in isolation. Together these lines form three orthogonal axes of open-vocabulary evaluation — open-set-aware scoring functions [18], static class-name quality [12], and (here) dynamic perturbation of the inference vocabulary itself; progress on one axis does not substitute for the others. Our protocol is designed to be re-runnable on future methods.

3 Audit Protocol

Frozen inference protocol. OpenCLIP [39] ViT-B/16 (OpenAI weights, QuickGELU); the 80 ImageNet prompt templates [8]; short side 336, sliding window 224, stride 112, mean logit fusion; temperature 40 softmax over sub-queries followed by per-class max pooling (the convention of Wang et al. [2]); identical for all methods and conditions. mIoU is computed over classes present in ground truth. No PAMR [41] or other post-processing. Under this protocol our SCLIP reproduction reaches 55.4 mIoU on VOC-21 with the official class-name file (paper: 59.1 with additional refinement), which we consider adequate for controlled comparisons since all conditions share the identical pipeline. Throughout the paper, differences and deltas quoted in the text are computed from unrounded run records; they may differ from differences of the displayed values (rounded to 0.1) by up to 0.1.

Table 1: VOC-21 (300 images). Official SCLIP vocabulary (background expanded to 26 sub-classes; synonym-enriched `person/tv`) vs. plain dataset names, identical protocol. The naming convention is worth 11.8–20.6 mIoU, comparable to or larger than the method gaps.

Vocabulary	SCLIP	ClearCLIP	MaskCLIP
Official (engineered)	**55.4**	53.6	43.6
Plain dataset names	34.8	**34.9**	31.8
Δ	-20.6	-18.7	-11.8

Axis 1: synonym substitution. For each class, candidate synonyms are WordNet [40] noun-synset lemmas filtered by CLIP text cosine similarity to the original name within $[0.70, 0.95]$; the highest-similarity candidate is chosen. Classes without valid candidates, and non-semantic classes (`background`, `unknown`, `other`), are never substituted. We replace 25/50/100% of substitutable classes with three random seeds at 50%. Ground-truth labels are unchanged.

Axis 2: granularity shift. Each class name is replaced by the first lemma of the first-sense direct WordNet hypernym (frozen rule, no manual correction), with the same exclusion list. The mapping table is part of the released protocol (Appendix D); merged-credit scoring (crediting predictions of a coarse name shared by several classes) is left to future work.

Axis 3: distractor injection. Distractors are drawn from the union of ADE-847 and PC-459 vocabularies minus the target vocabulary, filtered to CLIP text cosine ≤ 0.92 against all target classes (synonym guard), stratified into *near* (> 0.80), *mid* (0.65–0.80) strata, and injected at +50 and +200. Predictions of distractor classes count as errors for the ground-truth class.

Pre-registered repair hypotheses. Before running segmentation experiments we froze two kill criteria: (KC1) vocabulary-conditioned ZCA whitening of class embeddings must gain > 0.5 mIoU on large-vocabulary benchmarks across three methods; (KC2) vocabulary-conditioned statistics must beat global statistics in at least one regime. The pre-registered verdict was taken at shrinkage 0.5; a post-verdict sensitivity sweep over $\{0.1, 0.3, 0.7, 0.9\}$ (PC-459, ClearCLIP) confirmed the conclusion is not a shrinkage artifact.

4 Results

4.1 Naming conventions are worth more than method choice

Table 1 and Figure 1 show the single largest effect in the audit: replacing the engineered class-name file shipped with public code by the plain dataset names costs 11.8–20.6 mIoU. That class names matter in OVSS benchmarks was established by RENOVATE [12], which *renames* classes to better match masks; the measurement here is complementary—it quantifies, under a frozen protocol, how much of a reported training-free number is attributable to an undocumented name file rather than to the method. The method ordering is also regime-dependent: SCLIP leads ClearCLIP by 1.8 under the official file, the two are statistically tied under plain names (34.8 vs. 34.9, within subset-resampling noise), and ClearCLIP leads by 2.5 on PC-459. Since papers rarely document these files, cross-paper comparisons of training-free OVSS numbers largely compare vocabulary engineering, supporting the pre-registered naming-engineering and provenance claims.

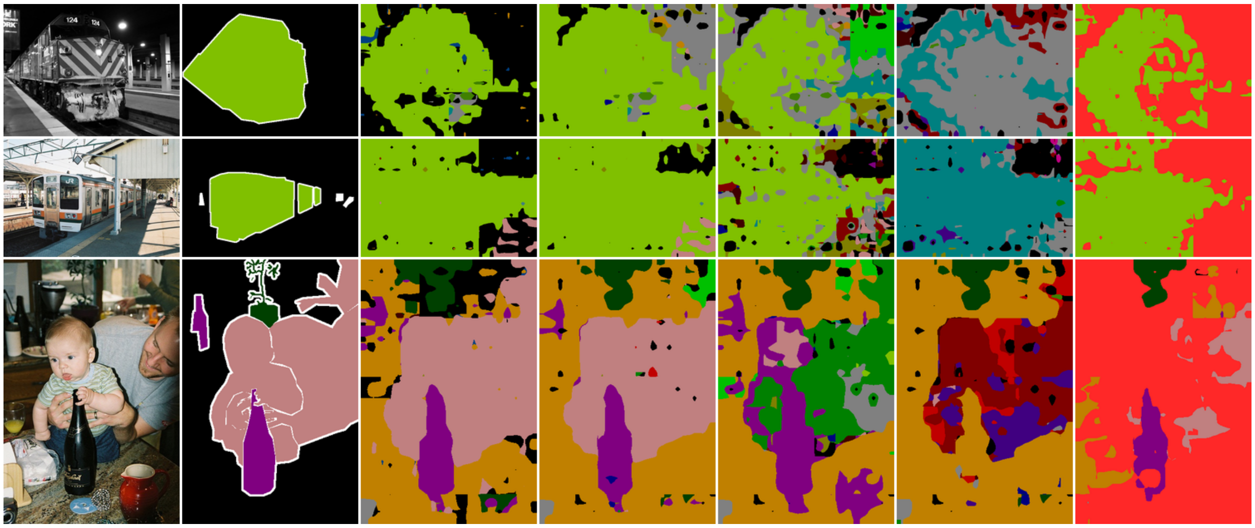

Figure 1: The same frozen SCLIP pipeline under five vocabularies (VOC-21 val). Columns: image, ground truth, official (engineered) names, plain dataset names, 100% synonym substitution, one-level WordNet hypernyms, plain +200 near distractors (distractor-class predictions in red). Only the list of class names changes between columns 3–7; every visible degradation is a vocabulary effect, not a model change.

Table 2: VOC-21 audit matrix (300 images, plain-name baseline, mIoU). Synonym substitution at rate r (seed 0; seeds 0/1/2 shown for 50%); one-level WordNet hypernym substitution (granularity). Labels are unchanged in all conditions. Single-subset values; the image-resampling noise floor is 3.1 mIoU (Fig. 2a), so differences below that are not interpreted.

Condition	SCLIP	ClearCLIP	MaskCLIP
plain (baseline)	34.8	34.9	31.8
syn 25%	34.1	32.9	29.6
syn 50% (s0/s1/s2)	34.4/31.3/30.7	33.3/32.8/31.2	29.4/29.3/27.7
syn 100%	30.6	29.0	26.3
hypernym (coarse)	17.0	17.0	16.9

4.2 Synonym and granularity axes

Synonym substitution costs 4–6 mIoU at full replacement, and the spread across three substitution seeds at 50% reaches 3.7 mIoU (Table 2); since all seeds are evaluated on the same fixed image subset, this spread reflects vocabulary choice alone, not image sampling (per-method mean±std across seeds: SCLIP 32.1 ± 2.0, ClearCLIP 32.4 ± 1.1, MaskCLIP 28.8 ± 1.0); on ADE-150 the one-level hypernym rule collapses performance from 13.1/14.7/10.2 to 2.5/3.8/1.9. Both axes use frozen, released rules, so these are reproducible lower bounds on user-facing robustness: a user who says *cattle* instead of *cow* or *motorcycle* instead of *motorbike* is not misusing an open-vocabulary system.

4.3 Distractor injection: mIoU rises while classification falls

Injecting 200 near-domain distractor names *raises* mIoU by 5–10 points (Table 3) while dropping matched classification accuracy by 18.5 points (single control cell: SCLIP encoder, VOC-21, +200

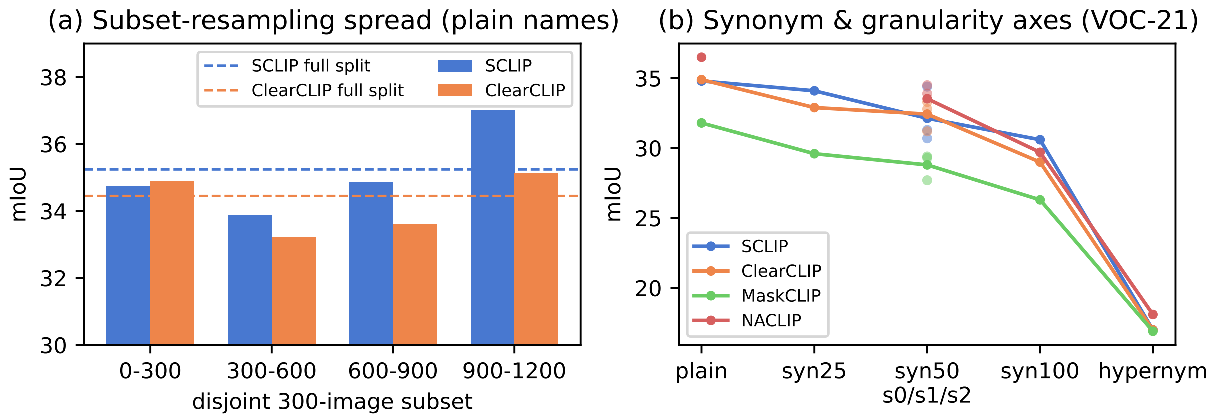

Figure 2: (a) Plain-name mIoU on four disjoint 300-image VOC-21 subsets vs. the full-split value (dashed): subset resampling alone spans up to 3.1 mIoU, our noise floor for method comparisons. (b) Synonym and granularity axes on VOC-21 for all four methods; dots show the three 50%-substitution seeds evaluated on the identical image subset.

Table 3: Distractor injection on VOC-21. Top: plain-name baseline; bottom: the same +200 near distractors added to the *engineered* official vocabulary (26 background sub-classes). mIoU is over ground-truth classes; the rightmost column is classification accuracy on ground-truth-box crops with the same vocabulary, templates, and pooling.

Condition	SCLIP	ClearCLIP	MaskCLIP	Cls. acc. (SCLIP enc.)
plain	34.8	34.9	31.8	72.8
+50 near	42.3	39.3	36.9	–
+200 near	44.5	40.9	37.1	54.3
+50 mid	41.5	38.2	36.0	–
official	55.4	53.6	43.6	–
official +200 near	52.5	51.6	41.9	–

near; ground-truth-box crops largely exclude background pixels, so this control shows the ranking behaviour of an object-centric protocol rather than an exactly matched task). The mechanism of the segmentation-side gain is background absorption: the plain `background` text embedding fails to capture background pixels, so distractor classes absorb false positives and boost the precision of ground-truth classes; mIoU over ground-truth classes then improves although every distractor prediction is an error. Two controls support the mechanism. First, when the same +200 distractors are added to the *engineered* official vocabulary—whose 26 background sub-classes already absorb background pixels—the gain reverses into a 1.7–2.9 point *drop* (Table 3, bottom): the anomalous gain exists only where background is under-modelled; once background is well modelled, vocabulary expansion behaves as expected and hurts. Second, under the standard all-class mIoU (distractor classes entering the average with IoU 0), the +200 near condition collapses to ∼4 mIoU: the direction of the vocabulary-expansion effect is itself a metric choice. A pre-registered control re-draws the 200 near distractors twice at random from the same frozen stratum (SCLIP/NACLIP, test-300): the all-class collapse is stable across draws (spread ≤0.4 mIoU), while the GT-present level varies by up to ∼4 mIoU (the deterministic first-200 draw is the harshest), so we treat the collapse as draw-independent and quote GT-present magnitudes to that tolerance. This also reconciles our

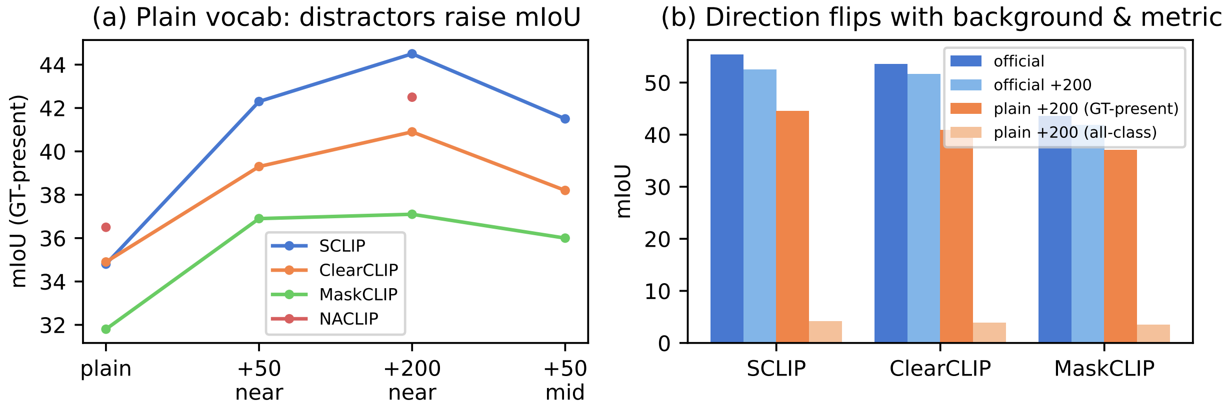

Figure 3: (a) Adding distractors to the plain vocabulary *raises* mIoU over ground-truth classes for all four methods. (b) The direction of the effect flips with background modelling and metric convention: on the engineered official vocabulary the same +200 distractors *lower* mIoU, and under the standard all-class mIoU the +200 condition collapses to $\sim$4.

result with reports that vocabulary expansion *hurts* [15]: with a well-modelled background or an all-class metric, expansion hurts as expected; the anomalous gain appears exactly in the common configuration—plain background name, mIoU over ground-truth classes—used by public training-free evaluations. A confusion-flow decomposition over all seven audited methods (Appendix C) makes this quantitative: the all-class drop is entirely a change of metric convention (ground-truth-present accuracy in fact rises), 64–72% of ground-truth pixel mass flows to the injected names — 83–88% of that flow from the background row, but 29–43% of *foreground* mass is also stolen, so absorption is the dominant, not the only, mechanism — inter-class confusion *decreases*, and the synonym axis shows the opposite signature (pure inter-class confusion, zero convention effect). A pre-registered capstone probe closes the repair question from above: pruning the vocabulary to the *ground-truth-present* classes (a presence oracle no filter can beat) leaves all-class mIoU at $\sim$4 on all three models tested — with a fixed 221-class denominator, 200 never-predicted classes contribute IoU 0, so the all-class figure is structurally capped near $(21/221)\times$ceiling for any method *including the oracle*, while its GT-present effect is small and mixed (-1.0 to $+6.5$). Scope: this holds under the fixed-denominator convention (our harness averages over all K classes, counting never-present, never-predicted classes as IoU 0); under a NaN-exclusion convention that drops 0/0 classes — as some standard toolchains do — oracle pruning would restore the denominator to the present classes and the "collapse" would disappear. The distractor collapse is thus unrepairable by presence-based vocabulary filtering under the fixed-denominator convention not because presence is unknowable but because the convention itself charges the denominator — which convention a toolchain uses silently decides whether expansion "hurts"; the open problem is metric design, not presence estimation. A pre-registered contamination curve makes the absorption side graded: injecting as few as 10 random absent names raises GT-present mIoU by +6 to +9 (SCLIP/NACLIP, saturating near +9 to +12 at 150), with random names absorbing *more* than near-distractors — the anomalous "expansion helps" regime begins at the very first injected name and is pool-dependent, so any two benchmarks with different vocabulary compositions already sit at different points of this curve. A pre-registered mechanism probe pins the absorption gain down as *background calibration in disguise*: meaningless pseudo-words reproduce half the random-name gain and 92% of filler-captured pixels are ground-truth background, but a single scalar background-logit boost (+0.02 in

Table 4: Text-side repairs under the frozen protocol (mIoU). ZCA uses shrinkage 0.5; global statistics from the ADE-847∪COCO-171 vocabulary. Whitening hurts in every cell (KC1 fails); vocabulary-conditioning edges out global statistics only on COCO-171, where both remain clearly below the unwhitened baseline.

Dataset	Method	none	center	ZCA (vocab)	ZCA (global)
PC-459	SCLIP	12.8	**13.7**	11.6	11.9
	ClearCLIP	**15.3**	15.2	13.2	13.7
	MaskCLIP	**9.6**	9.3	8.2	8.4
ADE-150	SCLIP	**13.1**	12.4	10.8	10.9
	ClearCLIP	**14.7**	13.9	12.7	12.9
	MaskCLIP	**10.2**	9.2	8.1	8.3
COCO-171	SCLIP	**22.9**	21.2	19.9	18.9
	ClearCLIP	**24.2**	23.1	21.7	21.4
	MaskCLIP	**17.4**	15.5	15.2	14.4

cosine units, no fillers at all) exceeds the full 50-name gain (47.5 vs 46.5 on SCLIP) — the anomalous expansion gain measures background under-calibration, and any absorption or negative-vocabulary claim should be benchmarked against this one-parameter control. A pre-registered check confirms the convention choice does not disturb *comparisons*: recomputing all seven methods under fixed-denominator, NaN-exclusion, and GT-present conventions leaves method-ranking Spearman at 1.0 on every axis. Two distinct facts should not be conflated here: the *oracle-pruned* ceiling of $\sim$4 is a convention product (NaN-exclusion would restore the denominator and dissolve it), whereas the *unpruned* collapse is real model behaviour under every convention — unpruned models fire on distractor names, so those classes are never 0/0 and NaN-exclusion does not rescue them. The convention decides what a number means, not which method wins. This shows, first, that vocabulary sensitivity of dense prediction is qualitatively different from classification, and second, that neither mIoU convention faithfully measures open-vocabulary segmentation quality under vocabulary expansion, supporting open-set-aware metrics [18].

4.4 Pre-registered repairs fail; geometry is not a proxy

Raw class embeddings exhibit the expected cone geometry (mean pairwise cosine 0.77 on A-847 to 0.82 on VOC-21; 10th-percentile nearest-neighbour margin shrinking from 0.078 to 0.037 as vocabularies grow), and ZCA whitening repairs every geometric diagnostic (A-847 margin 0.09→0.55, effective rank 154→309). Yet Table 4 shows ZCA whitening *reduces* mIoU in all 9 method×dataset cells and centering in 8 of 9 (the exception: centering helps SCLIP on PC-459 by +0.9); the post-verdict sensitivity sweep is monotone in shrinkage (PC-459/ClearCLIP: 11.6 at s=0.1 up to 14.7 at s=0.9 vs. 15.3 unwhitened), and vocabulary-conditioned statistics beat global ones only on COCO-171 (by 0.3–1.0), where both variants remain 2–4 points below the unwhitened baseline—so the conditioning advantage KC2 was designed to detect never materialises as a usable repair. The only consistently positive cases are small-vocabulary VOC-21 (+0.9 to +1.5 for ZCA on the 300-image plain split; archived in the artifact's run log, not tabulated in Table 4, whose scope is the mid-to-large regime where the verdict is taken). Two caveats bound the scope of this negative result: whitening is applied to text queries only, against un-whitened visual features, and we did not re-tune the logit temperature after whitening; joint or temperature-recalibrated variants might behave differently. The dissociation claim, however, needs only these cells: embedding separability improves monotonically while dense transfer degrades, so geometry diagnostics alone cannot certify

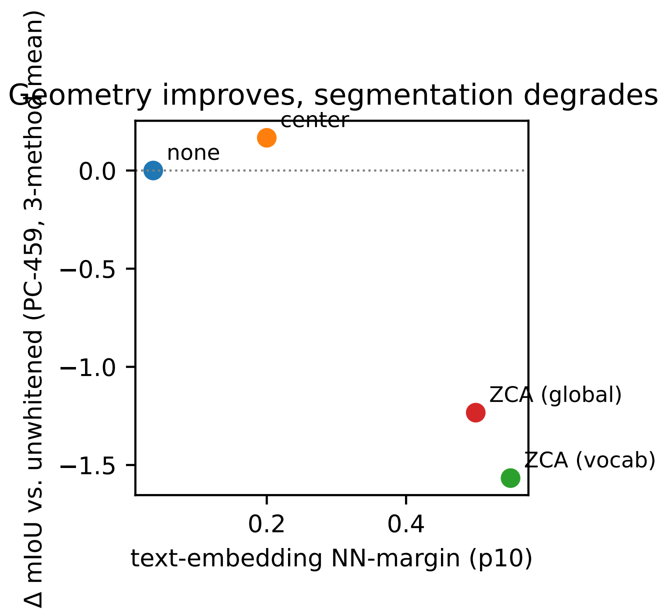

Figure 4: Text-side repairs on PC-459: each repair improves the embedding-space nearest-neighbour margin (x-axis) yet changes segmentation mIoU (y-axis, three-method mean) by at most +0.2 (centering) and down to −1.6 (ZCA, vocabulary-conditioned; largest single cell −2.1, ClearCLIP)—embedding geometry is not a proxy for dense segmentation quality.

a text-side repair.

Two further pre-registered failures. Two later probes sharpen the dissociation. (i) A frozen text-only vulnerability predictor (z-scored nearest-neighbour margin plus semantic drift to the plain names) was tested for ranking *vocabularies* against the archived benchmark mIoU values: median per-method Spearman is 0.03 on the full VOC suite (kill threshold 0.5) — the same class of geometry signal that ranks surgery *configurations* at a fixed vocabulary with $\rho = 0.89$ (Appendix B) carries no information across vocabularies at a fixed configuration. (ii) A transductive prediction-space repair (RECAL: name-conditioned additive logit debiasing toward a mass-flattening prior, estimated label-free from the evaluation stream, frozen before the run) reduces mIoU in every cell (SCLIP plain 34.8 → 27.6, syn100 30.6 → 23.3; NACLIP analogous): predicted class-mass imbalance under a perturbed vocabulary is dominated, in this setup, by legitimate scene statistics rather than name-conditional bias, so re-balancing it injects more error than it removes (verdict scoped to the single frozen prior strength α=0.5; no post-verdict sweep was run).

4.5 Can vocabulary engineering be automated? A constructive attempt

The audit's largest effect—the 12–21 point value of engineered vocabularies—raises an obvious constructive question: can the engineering be synthesised automatically for an arbitrary user vocabulary, training-free and text-side only? We pre-registered and tested such a method (Vocabulary-Adaptive Background Synthesis, VABS): build a candidate lexicon of scene/stuff words (COCO-Stuff ∪ ADE-150 names), remove candidates within cosine τ_{sim} of any target class, greedily select M diverse negatives (facility-location objective), append them as sub-queries of the background class, and fold their predictions into background. Hyper-parameters were selected on a 100-image

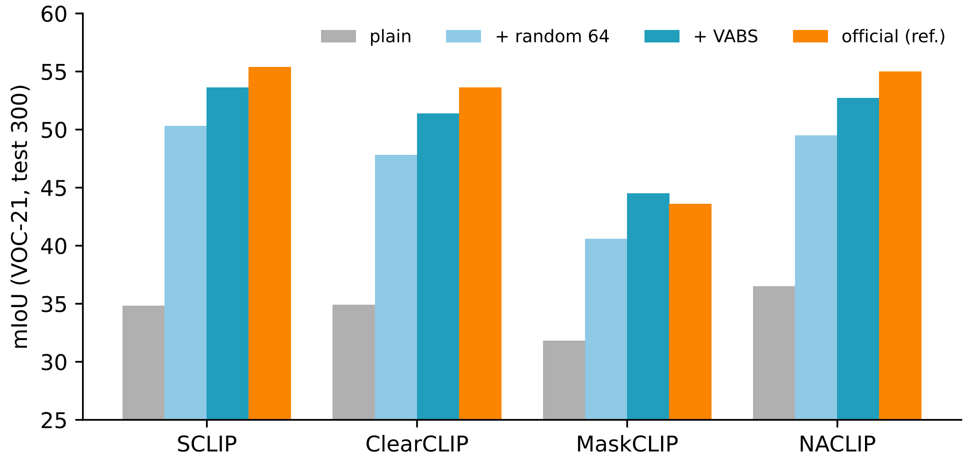

Figure 5: Automated background synthesis on VOC-21 (test 300 images). Synthesised negatives (VABS) close most of the plain-to-official gap for all four methods (official-vocabulary bars shown as reference); a matched random-word control already recovers the bulk of it, isolating the generic background-sink component.

Table 5: VABS on VOC-21 (test 300 images, GT-present mIoU). Automatic negatives recover most of the engineered-vocabulary gap and beat matched random negatives by +3.5 macro-average, without harming the official vocabulary. Right block: transfer to a COCO-Object-style protocol (80 things + background, stuff pixels remapped to background; 300-image subset — the full-val2017 plain baselines are 22.4/22.4 for SCLIP/NACLIP in the companion method paper).

	VOC-21					COCO-Object		
Method	plain	+rand. 64	+VABS	official	off.+VABS	plain	+rand. 64	+VABS
SCLIP	34.8	50.3	53.6	55.4	55.7	22.6	30.6	31.8
ClearCLIP	34.9	47.8	51.4	53.6	54.2	22.4	32.3	33.0
MaskCLIP	31.8	40.6	**44.5**	43.6	43.8	19.8	23.9	24.8
NACLIP	36.5	49.5	52.7	55.0	55.1	22.1	32.2	32.8

development set disjoint from the 300-image test set (M=64, τ_{sim}=0.9).

Macro gains are cheap and robust. Table 5: automatic negatives lift plain vocabularies by +12.7 to +18.8 mIoU (macro +16.1), reaching ≈96–102% of the official engineered vocabulary itself (on MaskCLIP, exceeding it); the official vocabulary is never harmed (+0.1 to +0.6). These gains are *not* the metric artifact of §4.3: there, injected classes compete with ground-truth classes and their predictions are errors under the all-class convention, whereas VABS folds every negative prediction back into the background class, so the improvement is realised background IoU under the unchanged 21-class metric. The effect partially transfers: on a COCO-Object-style protocol all four methods gain +5.0 to +10.7 (macro +8.9), but on PASCAL-Context-60, where the 59 classes already cover most stuff, the mechanism has no room (+0.5/+0.7 for SCLIP/NACLIP, indistinguishable from random negatives). One disclosure bounds the COCO-Object transfer: the candidate lexicon contains COCO-Stuff names, which that protocol remaps to background by construction; a leak-free ADE-only lexicon still gains +8.0/+9.8 (SCLIP/NACLIP) but no longer beats random words. Most of the gain is thus a generic *background-sink* effect: matched random negative words already recover +12.6 macro-average on VOC-21, and the selection advantage over random (+3.5 on VOC-21) shrinks to +0.6–+1.2 on COCO-Object. Figure 5 summarises the VOC-21 arms; NACLIP is included via the replication protocol of §5.1. VABS was also benchmarked against §4.3's one-parameter control, as that section demands of any absorption claim: a background-logit boost tuned on the labelled dev split reproduces 73–85% of VABS's text-side gain on VOC-21 (e.g. SCLIP 50.7 vs 53.5, matched folding, heldout-200), but the tuned scalar is a dataset-specific calibration constant — transplanted to Context-60 or ADE-150 it harms every cell (down to −9.6), and on protocols without a background channel it has nothing to attach to, whereas VABS transfers without labels or tuning. The verdict (VABS-DISTINCT) and per-cell numbers are archived in the artifact (W13-L5/L5t).

Per-class safety is not. A pre-registered per-class criterion (no target class may lose >3 IoU on ≥2 methods) failed on every variant we registered. The text-side filter cannot block negatives that are textually distant but visually overlapping: clothing words steal person pixels (−4.8 to −8.2 IoU on the three main-matrix methods, and up to −15.4 on an LPOSS-style base in the archived extension runs) and "screen" steals tvmonitor (−5.6 to −14.7 on four; near-destroyed at 2.1 IoU on the LPOSS-style base)—precisely the classes the official vocabulary protects with *positive*-side expansions ("person in shirt, . . . "). Two label-free visual repairs also failed their pre-registered criteria. A steal-rate guard (drop negatives claiming high-confidence target patches on unlabeled calibration images) is blind at high confidence—dense CLIP probabilities are so diffuse that *zero* person patches exceed a 0.5 softmax—and at low confidence the safe operating point removes so many candidates that the macro gain collapses to +2.5. Reassigning high-steal negatives as sub-queries of the stolen class (automating the official vocabulary's positive expansion) is worse: without labels, the targets-only argmax on background pixels is itself the pathology being repaired, so the diagnostic attaches "sky" to aeroplane and "water" to bottle (−28 IoU).

Why this is an audit finding. The gap between engineered and plain vocabularies thus has two parts: a large generic component (any background sink, automatable for free when the vocabulary under-covers the scene) and a per-class protective component that, in our attempts, required exactly the supervision signal (which pixels truly belong to class c) that a training-free segmenter cannot provide about itself. Engineered vocabularies appear to encode label-derived knowledge that our text-side and CLIP-confidence-based attempts do not recover; mechanisms with independent visual evidence (e.g. diffusion-derived negative prototypes [7]) remain untested here, but within the text-only regime, automating vocabulary engineering is not a hyper-parameter search.

4.6 Learned adaptation replays the failure on both sides

A final pre-registered attempt asked whether the missing robustness can be *learned* on the text side. We trained a lightweight residual adapter over the frozen text embeddings (vision tower untouched) plus a set of learnable background queries initialised from VABS negatives, on 8,000 unlabelled ADE20K training images (disjoint from every evaluation set), using region-pooled pseudo-labels from the strongest training-free configuration as teacher, a naming-invariance consistency loss over a WordNet variant pool (variants appearing in any evaluation vocabulary were excluded from training), an anchoring regulariser, and an anti-collapse margin term. Losses converged and the embedding effective rank stayed within 2% of frozen, yet every pre-registered criterion failed. (i) Learned *universal* background queries recover at most half of the vocabulary-adaptive VABS gain, for a structural reason: the best negatives for one vocabulary (sky, grass, water for VOC-21) are target classes of the training vocabulary, so no fixed query set can contain them. (ii) On held-out synonym vocabularies the invariance-trained adapter is *worse* than the frozen encoder (23.9

vs. 26.9 mIoU), and a direct variant→canonical normalisation objective is worse still (20.7); both variants also degrade the engineered-vocabulary regime (52.8→46.5–48.5 over VABS). Together with §4.4's linear repairs and §4.5's selection filters, three text-side mechanism families—linear transforms, learned adapters, and learned negative queries—now fail in the same direction: objectives that improve text-space geometry do not transfer to, and often damage, dense segmentation. A final pre-registered probe moved the same consistency objective to the *visual* side—a residual convolutional adapter over frozen dense patch features, trained on the same unlabelled images so that patch-level class distributions become invariant to synonym re-samplings of the vocabulary. Training was unstable (two runs hit a pre-registered feature-drift abort), and the last in-budget checkpoint degraded every regime: plain 36.5→30.2, held-out synonyms 29.7→25.3 (the same signature as the text-side failure), and official −0.9. With this, the naming-invariance objective has now failed on both sides of the embedding space, and five mechanism families—linear transforms, selection filters, learned text adapters, learned negative queries, and learned visual adapters—fail in the same direction. Two further pre-registered probes close the remaining repair spaces: a graded corpus-frequency predictor of per-name damage fails (median Spearman −0.22 over 757 per-name observations across nine models — within the synonym pool, rarer names are *not* more damaging, and the reversal is robust to swapping the wordfreq estimator for a CLIP-BPE merge-rank proxy, so the effect of §5.4 is binary non-canonical-name sensitivity rather than a frequency law), and a training-free discrete word-space repair (rewriting each incoming name to its highest-frequency WordNet alias under a CLIP-cosine guard) recovers 35–44% of non-canonical-control and searched (ANS) damage but violates its frozen no-harm criterion, costing 3–4 mIoU on already-canonical vocabularies: without a reliable canonicality signal — which neither text geometry nor corpus frequency provides — the rewriter cannot tell which names to leave alone. A third signal family, an LLM world-knowledge canonicality judge (frozen concept-blind prompt), passes no-harm but *worsens* every damage arm (−0.6 to −4.5 mIoU; −3.3 to −4.5 on the synonym and search arms): without the intended visual concept it resolves each name to its dominant word sense and rewrites legitimate synonyms across senses (verdict scoped to this judge configuration: a single LLM with a single frozen prompt, judgments archived before any run). Word-space repair thus fails under the three signal families we pre-registered; the missing ingredient is the intended concept, which a vocabulary-only repair by definition does not have. A pre-registered follow-up asked the prediction question directly: over 47 substituted COCO-Object classes (per-class IoU deltas, three hosts), neither the original-to-synonym cosine (rho +0.32 to +0.38, below the frozen +0.4 bar) nor the induced change in maximum inter-class cosine (rho $\approx$ 0) predicts per-class synonym damage — the text-space analogue of visual-side inter-class-correlation accounts does not carry first-order per-class signal, so which classes break under renaming cannot be read off first-order text geometry. Pre-registrations, checkpoints and run records are released with the artifact.

5 Discussion and Recommendations

Which decisions change. What is stable and what flips should be stated together. Stable: accounting conventions never re-rank methods (Spearman 1.0 on every axis, §4.3), and background-absorption corrections do not change any leaderboard. Not stable: the vocabulary *regime* does change who wins. Against the official-vocabulary ranking of Table 6, the plain and robust-mean rankings swap SCLIP/NACLIP and ProxyCLIP-/SC-CLIP-style (Kendall $\tau = 0.81$), the worst-case ranking swaps SCLIP above both ProxyCLIP- and SC-CLIP-style, and ranking by the naming-engineering gap (NEG) itself reorders more than half the pairs ($\tau = 0.24$; 8 of 21 pairwise leads flip, e.g. LPOSS-style has the best official score but nearly the smallest NEG). Under the searched worst case, the official leader collapses to the level of the least robust method (13.7 vs 13.4). A practitioner selecting a method for user-supplied vocabularies would therefore choose differently than the official leaderboard suggests — this, not the absolute numbers, is the decision-relevant content of the audit.

Mechanism exposure surfaces. The audited methods differ in *where* the vocabulary enters. In attention-surgery methods the class logits are direct inner products between patch features and the text embeddings, so every spatial decision is exposed to naming. In external-evidence methods the spatial structure comes from a vocabulary-independent source — most cleanly in the mask-generator anchor (§5.3), whose region proposals are pre-computed and never see a class name — which might suggest reduced naming sensitivity. Our anchors show the opposite: synonym fragility on the SAM-coupled and mask-generator anchors (5.8–12.4) matches or exceeds the attention-surgery family. This is consistent with the exposure living in the text–vision classification interface that all families share rather than in the spatial machinery they differ on (two external-evidence anchor families; small n); in the anchors we measured, fixing spatial structure did not shrink the naming attack surface.

For benchmark users. Report the exact class-name file; treat name files as part of the method. Report robustness under the released synonym and distractor perturbations (mean ± seed spread), not a single vocabulary. **For method designers.** Gains smaller than ∼3 mIoU on a single official vocabulary are within synonym-seed noise and should be treated as such. Background handling should be evaluated separately; distractor-injection response reveals it. **For metric design.** mIoU over ground-truth classes silently rewards background absorption under expanded vocabularies; open-vocabulary evaluation needs a metric that charges for distractor predictions, e.g. pixel-level open-set error decompositions.

5.1 Replication on a fourth method: NACLIP

As a generalization check beyond the initial trio, we re-ran the key audit cells on NACLIP [5], which augments k–k attention with a Gaussian neighbourhood prior (our re-implementation under the identical frozen protocol, without its extra refinements; same 300-image subsets as the main matrix). Every effect replicates: naming, official 55.0 vs. plain 36.5 (Δ18.5); synonym, 50% seeds 33.9/34.5/32.2 and 100% 29.7; granularity, VOC-21 36.5→18.1 and ADE-150 15.4→4.1; distractors, plain +200 near *rises* to 42.5 (+5.9) while official +200 near *drops* to 52.3 (−2.7); repairs on PC-459, ZCA 14.3→12.8 and centering 14.3→14.5 (a second small positive centering cell, matching the SCLIP exception). The audit's conclusions are thus not specific to the three methods of the main matrix, though all four remain final-block variants over the same text encoder.

5.2 Scale generalization: ViT-L/14

A pre-registered replication on OpenCLIP ViT-L/14 (OpenAI weights, identical protocol, VOC-21 test-300, SCLIP and NACLIP) shows that every audited effect *direction* transfers to the larger backbone, while magnitudes become method-heterogeneous. Synonym substitution at 100% costs 7.0 (SCLIP, 37.2→30.2) and 5.7 (NACLIP, 36.3→30.7); near-distractor injection leaves GT-present mIoU flat or rising (SCLIP −1.1, NACLIP +5.6) while all-class mIoU collapses to 3–4, reproducing the background-sink/metric decomposition. The naming effect, however, splits: NACLIP retains a large official-vs-plain gap (+14.3) whereas SCLIP's shrinks to +3.5 — on ViT-L the correlative-attention variant extracts far less from vocabulary engineering. Dense quality overall is lower than ViT-B for all cells, consistent with prior reports on dense CLIP at larger scales. We therefore state the audit's conclusions as direction-stable but magnitude-dependent on backbone–method pairing.

Table 6: Robust-mIoU benchmark (VOC-21, full dev-excluded split, 1349 images). *robust* is the mean and *worst* the minimum over {plain, syn50×3 seeds, syn100}; NEG = official−plain. The last column is all-class mIoU under 200 near distractors (GT-present in parentheses). All rows are our-protocol (re)implementations. Separate author-code anchor experiments (§5.3; ProxyCLIP, LPOSS, SC-CLIP, and Trident — not rows of this table) show smaller NEG under the ProxyCLIP and LPOSS pipelines but a matching NEG for SC-CLIP and Trident, so NEG *sizes* here are protocol-conditional. NEG is computed from unrounded run records and may differ from the difference of the displayed official and plain columns by up to 0.1.

Method	official	plain	robust	worst	NEG	distract.
MaskCLIP	44.6	32.8	29.7	26.8	11.8	3.5 (37.1)
SCLIP	57.2	35.6	33.5	31.8	21.6	4.3 (44.9)
ClearCLIP	55.7	34.8	32.5	29.8	20.9	3.9 (41.2)
NACLIP	56.9	36.9	34.0	30.7	20.0	4.1 (42.7)
ProxyCLIP-style	60.0	38.0	35.0	31.7	22.1	4.3 (44.8)
LPOSS-style	**60.5**	**40.8**	**37.3**	**32.7**	19.6	4.3 (45.5)
SC-CLIP-style	58.4	38.3	35.2	31.6	20.1	4.2 (44.1)

5.3 Generational generalization: 2024–25 visually-guided methods

A pre-registered extension asks whether the newest training-free generation — methods that inject an external self-supervised visual prior rather than editing CLIP's attention — has implicitly fixed naming sensitivity. We add three style-reimplementations to the unified protocol (disclosed as such, not reproductions): ProxyCLIP-style DINO [44] proxy attention, LPOSS-style DINO-affinity label propagation, and SC-CLIP-style anomaly-token restoration. On VOC-21 (full dev-excluded split, 1349 images) their official-vs-plain gaps are +22.1, +19.6 and +20.1 respectively — as large as the attention-surgery family — and all three collapse to all-class mIoU ≈4 under 200 near-distractors while GT-present mIoU stays at 44–46, reproducing the background-sink/metric decomposition. Vocabulary sensitivity is therefore a property of the shared CLIP text-matching interface, not of any particular dense-feature mechanism. One robustness-relevant difference does emerge: label propagation is the most vocabulary-robust mechanism on every perturbed aggregate (plain, mean-robust and worst-case); two pre-registered mechanism probes show this is a higher-plain-baseline effect rather than noise filtering — naming noise is graph-low-frequency, and transplanting the propagation operator onto the other six methods raises absolute mIoU but does not reduce the synonym drop. A cautionary methodological finding accompanies this: on a 300-image subset the *official*-vocabulary top spot went to proxy attention, but on the full dev-excluded split (1349 images) label propagation tops both rankings — the identity of the official-vocabulary leader is subset-sensitive at sub-0.5 mIoU margins, whereas the robust-aggregate ranking was stable across both evaluations, which itself argues for reporting perturbed aggregates.

Searched worst case (ANS). The random-synonym suite understates worst-case fragility. A pre-registered greedy search over per-class naming choices (WordNet candidates filtered to CLIP cosine $[0.70, 0.95]$; one coordinate pass; search and evaluation subsets disjoint) finds *lexicon-legal* vocabularies at held-out mIoU 13.4 (ClearCLIP) and 13.7 (LPOSS-style) — 15–17 mIoU *below* the worst random-suite member (28.3 / 31.0; random-suite figures are test-300 values, ANS figures heldout-200 — disjoint subsets, both disjoint from the search set) and with no search–held-out gap. The search is run independently per method, but a transfer matrix shows the found fragility is shared, not search adaptation: each method's searched vocabulary is nearly as damaging on the other method (cross-transfer 13.3 / 15.2 vs own 13.4 / 13.7). The bound is also robust to the search procedure itself: a pre-registered control re-runs the greedy pass on SCLIP under two random coordinate-visit orders alongside the alphabetical one (all three SCLIP searches run fresh for this control; the original searches used ClearCLIP and LPOSS-style); the three searches agree on 16/21 class choices and land within 1.4 held-out mIoU of each other (13.2–14.5 vs plain 34.7), so the reported worst case is not an artefact of the visit order. Notably, the most vocabulary-robust method in Table 6 (the LPOSS-style reimplementation) collapses to the same level as the least robust one: the label-propagation advantage vanishes under search. The full searched vocabularies, with a per-class plausibility annotation, are listed in Appendix A. We frame this as an upper-bound stress instrument rather than a typical-user scenario — the selected names are dictionary-sanctioned but often low-frequency senses (bicycle→wheel, person→soul, dog→frump) — but it shows that random perturbation suites measure average-case, not worst-case, naming robustness, and the searched axis is therefore reported separately in the benchmark. A pre-registered control decomposes what the search finds: replacing each ANS name with a *different* synonym from the same cosine pool, matched on CLIP token count (3 seeds), already costs 15.3 of the 21.0-point ANS drop on ClearCLIP — roughly 2× the random-suite drop. Most of the searched damage is therefore *non-canonical-name sensitivity*: any dictionary synonym that departs from the canonical class name hurts (random suites under-sample such names), with a genuine but modest residual adversarial component of roughly 5–7 points (control seeds span 18.5–19.7) on top. We deliberately do not phrase this as a graded frequency law: a pre-registered follow-up over 757 per-name damage observations across nine models found the correlation between corpus rarity and damage to be absent and slightly sign-reversed (median Spearman −0.22 with a wordfreq Zipf estimator, −0.12 with a CLIP-BPE merge-rank proxy closer to the training corpus), so the effect is binary — canonical vs. not — rather than monotone in frequency; the matched control above remains valid under this reading (matching on the candidate pool and token count matches non-canonicality, whether or not frequency grades it). The practical lesson for suite reuse: random synonym suites under-represent non-canonical names and therefore understate the tail by 2–3×; suites should either report a non-canonical stratum explicitly or, since no text-side signal we tested (geometry, frequency) can rank names, accompany average-case suites with an ANS-style search bound.

External anchor on unmodified author code. To rule out reimplementation artefacts, we ran the *unmodified* official ProxyCLIP release under its own protocol (mmseg pipeline, 2048×336 sliding inference, the authors' background thresholding, full VOC val), changing only the class-name file. The authors' name file scores 61.2 mIoU (matching the published number and within 1.2 of our style-reimplementation's official row); plain names score 47.3 (NEG +14.0); syn100 names score 41.2 (synonym drop 6.1). Naming engineering and synonym fragility therefore replicate on author code without any of our infrastructure — the smaller NEG relative to our protocol (+22.1) is partly explained by their background threshold absorbing part of the plain-vocabulary penalty, itself an instance of the background-modelling coupling documented in Section 4.3. A second author-code anchor, the unmodified official LPOSS release [16] (its own DINO-affinity label-propagation pipeline, 2048×448, full VOC val, again changing only the class-name list): authors' names 61.1 (published 61.2; our style-reimplementation's official row 60.5), plain 55.1 (NEG +6.0), syn100 47.7 (synonym drop 7.4). Direction replicates on both axes — and the synonym drop is *larger* on author code than our style-reimplementation's 4.3 — but the author-code NEG is far below our-protocol's 19.6: label propagation plus their pipeline absorbs much of the plain-vocabulary penalty, so leaderboard statements that depend on the *size* of the LPOSS naming-engineering effect are anchored to +6.0, not 19.6. A third anchor, the unmodified official SC-CLIP release [17] (its mmseg pipeline, 2048×336, full VOC val, again changing only the class-name file): authors' names 64.6 (matching the published 64.6 exactly), plain 43.0 (NEG +21.6), syn100 38.0 (synonym drop 5.0) — here the author-code NEG essentially *matches* our-protocol's +20.1 for the SC-CLIP-style row, because the official SC-CLIP name file carries the heaviest naming engineering in our set (a 26-item background expansion plus multi-alias foreground entries) and its pipeline, unlike ProxyCLIP's thresholding or LPOSS's propagation, has no mechanism absorbing the plain-name penalty. A fourth anchor extends the evidence to the SAM-coupled family, which none of our re-implementations cover: the unmodified official Trident [29] release (CLIP+DINO feature splicing with SAM [42] encoder aggregation and SAM refinement, its own protocol, full VOC val, again changing only the class-name file): authors' names 67.1 (matching the published 67.1), plain 47.4 (NEG +19.7), syn100 41.7 (synonym drop 5.8) — both axes replicate on a method whose visual pathway is qualitatively different from the dense-CLIP families, at a NEG size comparable to our-protocol values. Notably, Trident's pipeline includes SAM-based refinement, yet its NEG is undiminished: spatial refinement corrects boundaries, not naming — visual add-ons do not by themselves confer vocabulary robustness. The worst-case (ANS) axis also replicates on author code: the frozen search protocol run end-to-end on the official Trident release (query features recomputed exactly as its own initialization; disjoint frozen search/held-out subsets) drives held-out mIoU to 18.2 against plain 47.8 and syn100 40.7 — 22.5 below the random-synonym level, with no search–held-out shrinkage (search-100 was 20.4), so the searched fragility tail replicates end-to-end in an unmodified official pipeline: the worst-case regime is not specific to our protocol (demonstrated on one official codebase; single alphabetical search — the search-order control was not re-run on author code). The Trident ANS level (18.2) and our in-protocol ANS levels (13.4/13.7) come from different pipelines and subsets and are not directly comparable. The LPOSS ANS figures remain our-protocol-only. A fifth anchor adds venue freshness and a 2026 method: the official RF-CLIP [32] release (AAAI 2026, attention-redistribution family, CLIP-only ViT-B/16), changing only the class-name files (plus one memory workaround: the release's one-hot alias post-processing exhausts 24 GB with 145 alias queries, so we substituted a mathematically equivalent per-class max, regression-verified to reproduce the official VOC score bit-identically), reproduces its published scores on VOC (64.7 vs 64.8) and COCO-Object (36.8 vs 37.9, within tolerance) and shows the largest naming-engineering gap we measure on any author code: VOC NEG +23.8 (plain 41.0), with syn100 at 36.8 (drop 4.1); on COCO-Object NEG +4.9 and a synonym drop of 6.0. On VOC, the newest attention-surgery method is therefore the *most* dependent on engineered names among our anchors, not the least — 2026-era gains do not come with naming robustness. (On Context-60 the RF-CLIP official arm scored 1.57 above the published value, just outside our frozen 1.5 reproduction gate — likely our converted ground truth, as for Trident — so per protocol we draw no conclusions from that dataset for this method.) A sixth anchor covers the mask-generator family and the current training-free performance ceiling: the official CorrCLIP [30] release (ICCV 2025 oral; MetaCLIP [45] + DINO + the authors' pre-generated SAM2 [43] region masks) reproduces its published scores to within 0.1 (VOC 74.79 vs 74.8, COCO-Object 43.67 vs 43.7) and shows NEG +20.5 on VOC (plain 54.3) against +1.4 on COCO-Object, with synonym drops running opposite (5.8 vs 9.8) — the NEG/synonym dissociation observed on Trident replicates on a second SAM-family method. Because the region proposals are pre-computed and never see class names, every delta on this anchor is attributable to the text-side interface by construction. (Its Context-60 official arm lands 1.51 above the published value, marginally outside the same frozen gate — again likely our converted ground truth — so per protocol we draw no conclusions from that dataset for this method either; the authors' pre-generated Context masks cover 5104 of the 5105 validation images, so all Context arms for this anchor use the matching 5104-image list.) A seventh anchor covers the retrieval family that ReME opened and that our re-implementations do not reach: the official SCI-CLIP release [31] (training-free retrieval over a reference memory of region-pooled segment embeddings; in the released configuration we run, the bank is built from each benchmark's training split and regions come from SAM2 masks — a code-level description of the official pipeline as shipped, not of every configuration the paper reports). Its published CLIP ViT-B/16 scores reproduce closely on three of our four datasets — VOC 75.91 vs 75.9, COCO-Object 45.20 vs 45.2 (exact), ADE-150 29.83 vs 29.8 — while its Context-60 arm lands 1.56 above the published value, marginally outside the frozen 1.5 gate (again consistent with our converted Context ground truth), so per protocol we draw no conclusions from that dataset for this method. Changing only the class-name file (the frozen reference bank is reused across arms; retrieval is keyed on class indices, though the bank's stored label features derive from the official names — a residual coupling that favours the official arm and may inflate the perturbed-arm penalties, so the deltas on this anchor read as upper bounds): VOC plain 57.54 (NEG +18.4), syn100 52.23 (drop 5.3); COCO-Object plain 41.75 (NEG +3.5), syn100 32.37 (drop 9.4); ADE-150 plain 29.49 (NEG +0.3), syn100 15.29 (drop 14.2). This anchor thus reproduces every pattern in our set at once: a large VOC naming-engineering gap, the NEG/synonym dissociation across datasets (+18.4/5.3, +3.5/9.4, +0.3/14.2 — the same NEG-down/synonym-up ordering as Trident), and the largest synonym drop we measure on any author code (14.2 on ADE-150, an upper-bound reading given the bank coupling above): grounding class scores in retrieved *visual* references does not immunize the text-side interface, because the queries that address the bank are still CLIP name embeddings (one official implementation; family-level statements here are anchored to this single codebase). The spread between author-code and our-protocol NEG (ProxyCLIP +14.0 vs +22.1; LPOSS +6.0 vs 19.6; SC-CLIP +21.6 vs +20.1) shows that implementation and protocol details modulate effect *sizes* even where the direction replicates — and that, across the anchored methods, the modulation is method-dependent rather than a uniform reimplementation inflation.

A second round on Context-60 and COCO-Object (same suite minus official, which does not exist for these datasets *in our protocol*) confirms that synonym drops of 2–9 mIoU persist across all seven methods, though rank swaps there are minor — the label-propagation robustness advantage is largest on VOC. The SC-CLIP author-code anchor extends to COCO-Object, where the official release does ship an engineered name file: authors' names 37.7 (published 37.7, exact), plain 34.4 (NEG +3.3), syn100 27.9 (synonym drop 6.5, larger than its VOC drop of 5.0) — both axes replicate on author code on a second dataset. The NEG size, however, is 6× smaller than on VOC: naming-engineering gaps are strongly dataset-dependent, and VOC-scale NEG values must not be extrapolated to other benchmarks. The Trident anchor extends to COCO-Object with the same pattern: authors' names 41.1 (published 41.1, exact), plain 39.1 (NEG +2.0, 10× smaller than Trident's VOC +19.7), syn100 29.0 (synonym drop 10.1, far *larger* than its VOC drop of 5.8). The two axes therefore decouple on author code: a benchmark can have little naming-engineering headroom yet high synonym fragility — background headroom and naming sensitivity are different quantities and must be reported separately. A third Trident dataset sharpens this: on Context-60 (our 60-class conversion of the full annotations; the official arm scores 40.1 vs published 38.6, a gap of exactly 1.5 — at, but not beyond, the reproduction gate frozen before any Context run as "within 1.5 mIoU", so this dataset is retained while the 1.51–1.57 Context gaps of the other anchors fall outside it; label-conversion differences may contribute; all five arms share the same converted ground truth, so the NEG and synonym deltas are internally consistent — the conversion can shift absolute levels only) the naming-engineering gap is essentially zero (plain 39.9, NEG +0.2) while the synonym drop is the largest we measure on author code (12.4, to 27.5). The two axes therefore dissociate across the three datasets (n=3, an observation rather than a law): ordered by NEG size (+19.7, +2.0, +0.2), the synonym drop runs opposite (5.8, 10.1, 12.4) — consistent with the two axes being driven by different mechanisms.

5.4 Cross-family transfer: grounding-supervised detectors

All methods above share CLIP's contrastive text tower, so a natural objection is that naming fragility is an artefact of that one interface. A pre-registered transfer applies the same three axes to a different training paradigm: box-supervised open-vocabulary detectors converted to segmenters by prompting SAM with their boxes (Grounded-SAM-style harness [49]; detection scores arbitrate overlapping masks; unclaimed pixels are background). We audit Grounding DINO [48] (BERT [50] based phrase grounding) and OWLv2 [47] (CLIP-initialised text tower, detection-tuned) on VOC-21 and COCO-Object. Because the harness differs from the dense pipelines, all comparisons below are *within-detector deltas*; absolute mIoU values are not comparable across the two families. Three findings:

(1) Synonym fragility differs sharply within the detection family, and is not organised by text-encoder lineage. Grounding DINO collapses under legal synonyms: VOC 77.6→40.0 (−37.6; COCO-Object −28.3) — roughly 5× the CLIP-family drop — whereas OWLv2 degrades like the CLIP-surgery family (VOC −7.1, COCO-Object −12.9). Both detectors saw VOC/COCO categories in supervised pretraining, so the plain rows are quasi-in-distribution and these drops measure how tightly masks are bound to category *surface forms*, not zero-shot failure (the in-distribution advantage is shared, so it does not confound the contrast). An obvious reading is a text-encoder-lineage effect (BERT phrase grounding vs CLIP contrastive tower), and one prediction of that reading does hold: a worst-case vocabulary searched in CLIP text space (below) transfers strongly to the CLIP-tower detector while adding nothing on the BERT-grounded one (OWLv2 −31.8; Grounding DINO lands at its already-collapsed random-synonym level, −38.9 vs −37.6). A pre-registered third-model check, however, falsifies lineage as the organising variable: OWL-ViT v1 [46], which shares OWLv2's CLIP text tower, drops 19.0 under the same synonyms (64.0→45.0) — outside the frozen CLIP-tower band (≤15) and nearly in the BERT-grounded band (≥20). Since OWLv2 and OWL-ViT v1 differ mainly in training recipe (both share the CLIP tower; v2 adds self-training at scale), the contrast is consistent with a training-recipe effect rather than text-encoder pedigree; the Grounding DINO vs OWLv2 contrast stands as a model-level observation. A same-architecture natural experiment supports the recipe reading observationally: across the released MM-Grounding-DINO Swin-T ladder (identical architecture, training mixture varying from O365+GoldG to +GRIT to +GRIT+V3Det), the synonym drop moves by 10 mIoU (32.2 / 22.2 / 24.7) — the largest same-architecture spread we observe — with the web-grounding mixture the most robust; the pattern is not monotone in data volume in this three-point sweep (and each tier entangles composition with scale), so it demonstrates that the recipe matters without identifying which ingredient or licensing a data-scale story. A second attempted third model triggered the pre-registered infeasibility clause (plain GT-present < 35, frozen before any synonym result was seen) and is itself an interface finding: MDETR (RoBERTa grounding, no presence head) grounds *every* caption somewhere in every image, so per-class querying floods the image with boxes (3.4 mIoU regardless of vocabulary) — grounding without a presence decision cannot support vocabulary-conditional segmentation in this harness.

(2) The searched worst case transfers across paradigms. The ANS vocabulary searched on ClearCLIP (§5.3) costs OWLv2 31.8 mIoU (73.4→41.6 held-out) — 4.5× its random-synonym drop — despite OWLv2 never being touched by the search. Random suites therefore understate worst-case fragility across training paradigms, not just within the searched family. A pre-registered token-matched control bounds what transfers: matched non-canonical synonyms drawn from the same candidate pool cost OWLv2 25.3 of the 31.8-point ANS drop (80%; 3 seeds, 44.1–50.3 vs plain 73.4). The cross-paradigm transfer is therefore mostly shared *non-canonical-name sensitivity* — itself a cross-family phenomenon random suites underestimate by ∼3× — with a modest search-

specific increment (∼6.5 mIoU).

(3) No foreground absorption in the detection family. Under 200 injected near-distractors, GT-present mIoU stays essentially flat (Grounding DINO −4.3, OWLv2 −1.3, vs. a +4.6 to +9.8 *rise* for the CLIP family). This is not mere dilution: the detectors do fire on the injected names (17.8% / 23.6% of pixels are claimed by distractor classes), but ∼92% of those pixels lie on ground-truth background, and only 4.5% / 8.5% of foreground mass is stolen (CLIP family: 29–43%). The explicit per-query presence decision produces genuine open-set false positives on background regions while protecting foreground pixels — corroborating the two-ledger decomposition of Appendix C from a second architecture family, where the all-class collapse (≈7) is again the metric-convention term.

Scope: a box→SAM harness rather than native segmentation (synonym drops could partly reflect boxes falling below the fixed detection threshold rather than degraded matching; a threshold sweep was not run); one seed; test-300/held-out-200 subsets vs the dev-excluded full split of Table 6; Grounding DINO's 256-token text limit requires 40-class query chunks on the distractor vocabulary, OWLv2 queries classes independently.

A pre-registered native-metric check removes the harness confound entirely: evaluating OWLv2 and YOLO-World as *detectors* (official COCO `pycocotools` mAP, val2017 first 1000 images, only the class-name strings changed) reproduces both signatures. Full-synonym vocabularies cost ∼30% of mAP on both models (OWLv2 48.4→36.2/31.1 under two frozen synonym vocabularies; YOLO-World-L 46.8→35.3/30.0), so synonym fragility is a property of the vision–language interface, not of our box→SAM conversion or of mIoU conventions. Injecting 200 near-distractor queries, by contrast, barely moves native AP (−1.2 / −2.6 mAP): per-query decoding has no shared softmax or background ledger to absorb, so the segmentation-side distractor collapse has no detection analogue — corroborating, from a third angle, that the collapse is a metric-convention phenomenon rather than lost recognition. A decomposition at IoU 0.5 attributes the synonym drop chiefly to outright recall loss (GT recall 78.5%→58.1%; 21.5% of objects are no longer detected at any score) rather than score reordering. Two further pre-registered detection-side couplings came back *null* and we report them as such: letting original names and synonyms coexist as 160 queries neither degrades mAP (−0.7) nor produces duplicate boxes (0.2% at IoU≥0.9), and a cross-name NMS merge changes nothing (−0.2) — the per-query interface is robust to vocabulary redundancy, which is precisely why these findings live here as an audit extension rather than as a separate detection study (frozen go/no-go criteria: the phenomenon-existence condition passed at 2× margin, the detection-specific-coupling condition failed both thresholds).

5.5 A cross-lingual probe: a Spanish case study

Real vocabularies are not always English. A pre-registered quick test (fixed dictionary translations of the VOC-21 names, frozen before any run) found Spanish structured and Chinese a floor collapse, so the Spanish axis was promoted to the full matrix (VOC test-300, 9 models); we scope this as a single-language probe rather than a general cross-lingual axis — broader typology is untested. Translations are fixed common dictionary renderings frozen before any run (provenance in the released vocabulary files). The profile is a family split, not a method ranking: all seven dense CLIP-family methods — across three method generations — retain a tightly clustered 52–62% of their plain mIoU (16.5–23.4 absolute), OWLv2 retains 79% (72.5→57.3; test-300 baseline), and Grounding DINO retains 18% (77.6→14.1), under default translations; the per-class best-of-3 oracle raises Grounding DINO only to 25% retention (19.7 vs plain 77.6), so the family split is not a translation-choice artifact. The axis decouples from the synonym axis in an informative way: the dense family's synonym-robustness spread collapses to a uniform band under translation, because the cross-lingual response is set by the shared CLIP text tower's multilingual capability rather than by any dense-inference design choice; the detector contrast (OWLv2 most robust on both axes, Grounding DINO most fragile on both) survives (one model per detector family; the contrast is a model-level observation). Note this does not resurrect the lineage hypothesis falsified in §5.4: multilingual capability tracks the text tower almost by construction (CLIP towers saw multilingual alt-text; BERT-grounding saw English grounding data), whereas *synonym* robustness demonstrably does not track lineage — the two statements concern different mechanisms and are compatible. Chinese collapses everywhere (<10 mIoU; for OWLv2 the translated queries do not even fit the 16-token CLIP context and require truncation — the boundary is partly tokenizer representability, disclosed as such), so it is reported as a boundary note rather than an axis; pre-registered artifact controls (bare-name queries without the English template, and shortened 2-character forms) recover nothing (< 1 mIoU gain, mostly negative), so the collapse is genuine multilingual incapacity rather than a prompt-interface artifact. For Spanish, by contrast, a pre-registered best-of-3 translation control (two frozen dictionary alternates per class, per-class oracle — an upper bound biased *against* this reading) recovers 86% of plain-English mIoU on SCLIP (30.0 vs 34.8; single translations 14.7–20.7), crossing the frozen translation-choice band, and 84% on OWLv2 (62.0 vs 73.4; heldout-200 baseline, hence the difference from the 72.5 test-300 figure above), same direction but just under the 85% line: the 52–62% single-translation figures largely measure sensitivity to *which* legal translation is chosen — the naming-choice fragility of §4.2 re-appearing in another language — rather than a Spanish capability deficit (the family split across translations survives). Chinese, per the artifact controls above, remains a genuine capability boundary. A pre-registered two-language extension (German, Russian; k=3 frozen translations each) finds the same structure at language-dependent magnitude: oracle-minus-default retention gaps of 14.0 (German) and 10.9 (Russian) points on SCLIP and 12.7 (German) on OWLv2, while Russian on OWLv2 collapses to a floor (5.9% retention) — a second non-Latin-script data point alongside the Chinese boundary (for Russian we ran no artifact controls analogous to H4's, so tokenizer-representability artifact and genuine incapacity are not separated there). In every language we tested (three languages, k=3 frozen translations — a lower bound on the oracle gap), translation choice is a substantial free variable, and single-translation cross-lingual scores should be read with an oracle-gap control; broader typology is future work.

6 Limitations

Our audit covers three final-block CLIP variants under one backbone (with a two-method ViT-L/14 direction replication, §5.2); because they share the text encoder, templates, and pooling, their correlated response to vocabulary perturbation is partly by construction, and all method-general statements are scoped to this family (§5.1 extends it to NACLIP, a fourth member). The generational extension (§5.3) adds ProxyCLIP-, LPOSS- and SC-CLIP-style reimplementations plus author-code ProxyCLIP, LPOSS, and SC-CLIP anchors, and §5.4 transfers the protocol to two box-supervised grounding detectors through a box→SAM harness; SAM-based pipelines (e.g. Trident-class refiners) as *audit subjects* remain open (the protocol applies unchanged). The discrete word-space rewriter's partial success against searched and non-canonical attacks (35–44% recovery) indicates a repair direction that a reliable canonicality detector would unlock, though none of the signals we tested provides one. A-847 was unavailable at audit time; PC-459 serves as the large-vocabulary regime. Evaluations use fixed, released 300–500-image validation subsets per condition for tractability of the ∼150-run matrix. Two calibrations bound the subset error: on the full VOC-21 validation split (all 1449 images; the dev-excluded 1349-image variant used elsewhere differs by at most 0.4) the naming effect persists at full strength (plain 35.2/34.5/32.5 vs. official

56.9/55.4/44.4 for SCLIP/ClearCLIP/MaskCLIP, Δ 11.9–21.7), and re-evaluating the plain condition on four disjoint 300-image subsets gives a subset-resampling spread of up to 3.1 mIoU (SCLIP) and 1.9 (ClearCLIP). Effects of this order or smaller (including the 0.1-point VOC-21 plain-name difference between SCLIP and ClearCLIP) are treated as noise throughout. The frozen perturbation thresholds (synonym similarity window, distractor guard and strata boundaries) were fixed before segmentation runs but not themselves ablated; first-sense WordNet hypernym mapping can produce non-visual parents, so the granularity numbers are lower bounds under a fixed automatic rule rather than human-validated coarsenings. The classification control is a single cell and uses object crops (see §4.3). The remaining gap of our official-vocabulary reproduction to published numbers (full-split 56.9 vs. 59.1 for SCLIP) is consistent with our protocol omitting PAMR post-processing. A pre-registered check bounds this axis: adding PAMR refinement to SCLIP/NACLIP (VOC test-300, three vocabulary conditions) leaves the synonym drop unchanged (5.5 vs 5.4 mean) and *amplifies* the naming-engineering gap by 15% (19.6 $\rightarrow$ 22.4) — spatial post-processing helps the engineered vocabulary more than the plain one, so the audited magnitudes are not artifacts of omitting refinement (multi-scale and full published stacks remain unbounded). Scoop-check against in-review venues (OpenReview) was not possible; novelty claims are made against the published literature.

7 Conclusion

Treating the inference vocabulary as a controlled experimental variable reveals that reported numbers for final-block training-free OVSS methods are strongly shaped by vocabulary artifacts: name engineering comparable to or larger than method differences, synonym noise larger than typical claimed gains, a mIoU protocol whose response to vocabulary expansion is an artifact of background modelling and metric convention, and text-side geometric repairs that help every diagnostic except actual segmentation. A constructive attempt to automate vocabulary engineering (§4.5) recovers most of its macro-mIoU value with synthesised background negatives, but its per-class protective component resisted every label-free mechanism we pre-registered, indicating that engineered vocabularies encode supervision that current label-free machinery—linear, selected, or learned, on either the text or the visual side (§4.6)—cannot reconstruct. We release the protocol and harness so that vocabulary robustness can become a standard reported quantity for open-vocabulary segmentation.

A ANS Searched Vocabularies

Table 7 lists every class whose name the greedy search replaced, for both searched methods, with a coarse plausibility annotation: *plausible* = a name an ordinary user might type for the class; *legal-but-odd* = a WordNet-sanctioned synonym of a rarer sense that a user is unlikely to intend. Most of the damage comes from legal-but-odd choices, supporting the “upper-bound stress instrument” framing; a handful (e.g. *lounge* for sofa, *soul/mortal* for person) are plausible user inputs, so the searched axis is not purely adversarial.

B Head-Level Decomposition of Final-Block Surgery

The audited methods differ only in the final attention block, inviting a mechanistic question: *where* inside that block does dense quality live? We decomposed the final-block output into per-head contributions (via the output projection’s column blocks) for four attention flavours (vanilla q–k, q–q, k–k, identity/value-only) and evaluated each single head and head subset under the unified

Table 7: ANS replacements (VOC-21). Unlisted classes kept their plain name.

class	ClearCLIP choice	LPOSS-style choice	plausibility
bicycle	wheel	wheel	legal-but-odd
bird	skirt	skirt	legal-but-odd
boat	gravy holder	gravy holder	legal-but-odd
bottle	bottleful	nursing bottle	legal-but-odd
bus	charabanc	charabanc	legal-but-odd
car	railroad car	railroad car	legal-but-odd
cat	guy	guy	legal-but-odd
chair	chairwoman	chairwoman	legal-but-odd
dog	frump	frump	legal-but-odd
horse	sawhorse	sawhorse	legal-but-odd
person	soul	mortal	plausible
sofa	lounge	lounge	plausible
train	caravan	caravan	legal-but-odd

protocol (VOC-21, 100 development images, official vocabulary). Two descriptive findings and one negative result. (i) The head spectrum is extremely uneven (single-head spans of 13.6–21.6 mIoU) and *flavour-invariant*: head rankings correlate at Spearman 0.92–0.96 across all four flavours, with heads {3, 9, 11} carrying dense semantics in every case—surgery methods appear to amplify one fixed head structure rather than create new ones. (ii) A label-free alignment margin (mean top1−top2 text similarity over patches) predicts the segmentation quality of 20 surgery configurations (4 flavours $\times$ exit layers 8–12) at Spearman 0.89 without any annotation, though restricting the margin to the top-3 heads weakens it to 0.63, so the predictive signal lives at the whole-output level, not specifically in the head structure. (iii) Head *selection* is not a method: oracle subsets beat all-heads by only +0.4 to +0.6 on surgery flavours (+2.8 on vanilla), and label-free selection loses up to −2.6; the surgery has already harvested the fixable component. All 20 configurations and per-head records are in the artifact release; sample sizes (100 images, 20 configurations, one backbone) bound these as within-distribution observations.

Two pre-registered follow-up probes sharpen the picture with oracle upper bounds. (iv) Decomposing the similarity itself into per-head subspace terms $\sum_h w_h \langle P_h t, P_h v\rangle$ (with P_h the projector onto head h's output subspace, linearised through the final layer norm) does not help even with GT-fitted weights: the held-out gain is +1.1 mIoU on vanilla attention (below the +2.8 oracle head-selection bound) and −5.4 on q–q; a uniform-weight decomposition loses 2–6 mIoU against the plain inner product, indicating that cross-subspace interference terms carry signal that any per-head decomposition discards. (v) Routing *configurations by region* has a large oracle ceiling but no label-free access to it: choosing the best of the 20 configurations per SAM region with GT reaches 68.2 mIoU versus 53.2 for the best single configuration (+14.9; an image-level oracle gets only +4.3), yet every label-free gate we pre-registered (region margin, confidence, cross-configuration agreement over the margin-selected top-4 pool) lands *below* the best single configuration (38.2–46.7)—consistent with (ii): the margin signal predicts quality globally but not at region granularity. Both probes were killed by their pre-registered criteria and are reported as mechanism-level negative results with their oracle gaps. A follow-up asked whether a *small amount of supervision* unlocks the routing ceiling: a ridge router over region margin, confidence, entropy, size and cross-configuration agreement features, fit on 50 labelled images, transfers negatively to held-out images (53.4 vs. 54.0 for the best member) and degenerates on a disjoint 300-image split to always selecting the best single configuration (+0.0). The routing signal is not present in this family of region-level confidence statistics even with labels; accessing the oracle gap appears to require qualitatively different evidence. Two further pre-registered probes closed the remaining signal families we could test. (vi) *Cross-configuration consistency* routing (route each region to the configuration agreeing with the 8-pool majority vote, optionally gated by DINO boundary crispness) lands below the best single configuration on both VOC (46.8 vs. 54.2) and Context-60 (21.9 vs. 26.4), and a margin control performs equally — the third signal family (after margin and few-label supervision) that fails to access the routing oracle; notably, an 8-configuration pool restricted to exit layers 11–12 has an oracle ceiling of only +2.4, showing most of the +14.9 headroom lives in early-exit configurations that are individually too weak for any realistic gate to select. (vii) A *conformal prediction-set* wrapper over region scores (split conformal on 500 COCO-calibrated regions, nonconformity from vocabulary-ensemble mean and variance) fails to transfer across vocabulary sizes: at a 90% target, coverage drops to 59.7% on VOC-21 with prediction sets covering 72% of the vocabulary, because softmax-based scores are not exchangeable across vocabularies of different cardinality (within-distribution calibration on Context-60 is valid at 92.3%, but that weaker per-dataset form was outside the pre-registered claim). (viii) Two pre-registered probes of the *propagation-smoothing* hypothesis — that LPOSS-style label propagation is robust because it low-pass-filters text-side naming noise over the DINO affinity graph — both failed. A graph-spectral decomposition shows that vocabulary-perturbation logit noise is 97% *low*-frequency on the affinity graph (high-band energy fraction 0.029), i.e. naming noise is spatially class-coherent, leaving spatial or graph smoothing nothing to filter; and transplanting the propagation operator onto the six other methods raises absolute mIoU on every one of them (plain +1.7 to +12.8) while *increasing* the synonym drop on all six. The robustness advantage of label propagation in Table 6 is therefore a higher-plain-baseline effect, not noise filtering — a further link in the chain that text-side naming noise is not repairable by any downstream spatial mechanism. (ix) A pre-registered *presence-gating* wrapper (rank-based gate on top-3 SAM-region-pooled class probabilities, frozen threshold) fails to repair distractor collapse: under 200 near-distractors the gate retains half the vocabulary (precision 0.24 at recall 0.99) and leaves all-class mIoU at $\approx$4 on both ClearCLIP and NACLIP — region-pooled softmax mass does not separate present from absent vocabulary items at this granularity. A second, pre-registered signal family — raw-cosine margins against VABS-selected negatives combined with a top-region winner-consistency score, explicitly distinct from the killed rank gate — mildly *helps* on official and plain vocabularies (+0.2 to +1.5) but equally fails the distractor repair criterion (all-class mIoU 3.8/4.2 vs. a pre-registered bar of 20; precision 0.31–0.35 at recall 0.98): image-level presence of a vocabulary item is not recoverable from either softmax-rank or raw-margin region statistics. Scope of this verdict: two signal families, two methods (ClearCLIP/NACLIP), and the high-recall operating regime that protecting ground-truth classes requires (recall ≥ 0.98 under the frozen gates); we did not sweep the precision–recall curve for a post-hoc operating point, as both pre-registrations froze their thresholds. Retrospective note: the GT-presence oracle probe (§4.3) later showed that the all-class mIoU bar of 20 is structurally unreachable under our fixed-denominator convention — even a perfect presence gate is capped near 4 — so the mIoU criterion in these two pre-registrations was invalid as frozen; the retirement of presence gating rests on the precision evidence alone (0.24–0.35 at the recall a gate must operate at to protect ground-truth classes — at that precision roughly half the injected vocabulary survives the gate regardless of the metric convention). Within that scope presence gating was retired for this project.

B.1 Index of pre-registered negative results

Table 8 indexes every pre-registered repair or predictor hypothesis retired during this project, with its signal family, frozen criterion, and outcome; full records accompany the artifact.

Hypothesis	Signal family	Frozen criterion	Outcome
ZCA/shrinkage whitening, centering	text geometry	no regime harmed	harms almost every regime
Learned text adapter (invariance / normalisation)	learned text	held-out synonym gain	worse than frozen encoder
Learned universal background queries	learned text	match VABS gain	≤half; structurally capped
Learned visual adapter	learned visual	no regime harmed	degrades every regime
Region routing (label-free / few-label / consistency)	CLIP confidence	approach oracle +14.9	none approach
Conformal VocabUQ	CLIP confidence	cross-dataset coverage	fails transfer
Presence gating (rank; margin+consistency)	CLIP region stats	precision at recall ≥0.98	0.24–0.35; retired†
Propagation smoothing transplant	spatial graph	reduce synonym drop	increases it on all six
Graded frequency predictor	corpus frequency	median $\rho \geq 0.5$	−0.22 (sign reversed)
Text-geometry vocabulary ranking	text geometry	median $\rho \geq 0.5$	0.03
Discrete canonicalisation rewrite	word space	no-harm on plain	−3 to −4
LLM canonicality judge (single config.)	LLM world knowledge	no-harm + recovery	passes no-harm, negative recovery
Transductive logit debiasing (α=0.5)	prediction space	no regime harmed	large negative everywhere
OWLv2 box shield (distractor axis)	external detector	all-class recovery	4.23→4.27
Text-encoder-lineage banding	model family	third model in band	falsified (OWL-ViT v1)
Alias-diversity class-level law	training corpus	class-level $\rho \geq 0.25$	0.107
Absent-leak class-level predictor	pixel-flow accounting	class-level $\rho \geq 0.25$	−0.20 (pooled contrast real)
GT-presence oracle repair	ground-truth presence	restore all-class mIoU	capped at ∼4 (metric, not presence)
Contamination-curve protocol	effective vocabulary	GT-present drop ≥ 3	rises +6 to +9 from 10 names
Metric-convention re-ranking	accounting convention	ranking change on any axis	Spearman 1.0 everywhere
Pseudo-word free-lunch expansion	non-semantic fillers	≥80% of real-noun gain	52%; beaten by one scalar
Detector-granularity recall law	WordNet depth	$\rho \leq -0.3$	+0.17 (sign reversed)

Table 8: Pre-registered negative results (kill table). †: the all-class mIoU bar in the presence-gating pre-registrations was later shown structurally unreachable (App. B); the retirement rests on the precision evidence.

C Confusion-Flow Decomposition of Perturbation Damage

To make the two-factor account of distractor collapse (§4.3) quantitative, we store the full ground-truth-row confusion matrix for all seven methods under plain, syn100 and +200-near vocabularies (VOC-21 test-300 and COCO-171 val-300; this mechanism appendix uses the fixed subsets rather than the dev-excluded full split of the main leaderboard — flow *proportions* are aggregate pixel statistics over ∼70M pixels per cell and are far less subset-sensitive than per-class mIoU) and maintain two separate ledgers, which we deliberately do not mix because they have different units. *Pixel-flow ledger* (fractions of total ground-truth pixel mass; the three terms plus the correct-mass change sum to zero by construction): flow into the background class, flow into injected distractor names (steal), and flow into other ground-truth classes (inter-class confusion). *Metric ledger* (mIoU points): the all-class drop versus the change in GT-present accuracy. Both ledgers are qualitatively identical across all seven methods and both datasets. On the distractor axis the metric ledger shows the all-class drop is entirely a change of metric convention rather than an accuracy loss: GT-present mIoU *rises* on VOC (+4.6 to +9.8 across methods) while all-class mIoU drops 28–37 points (the convention gap on the perturbed run is 34–41 points; COCO: GT-present flat within ±0.3, convention gap 9–15). This does not mean the collapse is harmless — every distractor prediction is a real error under an open-vocabulary reading; it means neither mIoU convention prices those errors faithfully, consistent with §4.3. The pixel-flow ledger locates the errors: 64–72% (VOC) / 15–21% (COCO) of ground-truth pixel mass moves to distractor names. On VOC, 83–88% of that flow comes from the background row (per-method range across the seven methods), quantitatively confirming the background-absorption mechanism; but 29–43% of *foreground* pixel mass is also stolen, so absorption is the dominant, not the only, mechanism (COCO-171 has no background class, so its steal is foreground by construction). Flow into the background class is essentially unchanged and inter-class confusion *decreases*. On the synonym axis both ledgers are entirely different: zero steal, zero background flow, no convention gap — the synonym drop is pure inter-class confusion and hence a genuine accuracy loss. GT-present method rankings are identical to all-class rankings on every axis and dataset (Spearman 1.0), so the decomposition changes interpretation, not leaderboards; we therefore report it as protocol clarification rather than a new metric.

D Released Artifacts and Run Provenance

The release accompanying this paper (a public code repository, linked from the arXiv page once posted) contains: (i) all perturbed vocabulary files (synonym 25/50/100% $\times$ 3 seeds, hypernym mappings for VOC-21 and ADE-150, distractor pools and injected vocabularies per stratum and size, and the exclusion list {`background`, `unknown`, `other`}); (ii) the perturbation scripts with frozen thresholds; (iii) the evaluation harness implementing the uniform inference protocol; (iv) the fixed image-subset lists; and (v) one JSON record per evaluation run (method, dataset, vocabulary file, whitening configuration, subset, per-class IoU, both mIoU conventions), from which every table cell in this paper is reproduced by a lookup script. An early internal run series contained a protocol bug in which the non-semantic `background` class could be synonym-substituted; those runs were archived, the exclusion rule was added to the frozen protocol, and all perturbed vocabulary files were regenerated from the corrected pipeline. Because pool construction and sampling share one code path, the regenerated distractor sets are not byte-identical to the archived ones, which shifts distractor cells slightly (e.g. SCLIP +200 near: archived 45.4 vs. corrected 44.5); all synonym and distractor conditions were re-run on the regenerated files. Every number reported in this paper, including the full-split and subset-resampling calibrations, comes from a completed, on-disk run record of the corrected series. The constructive experiments of §4.5 follow the same discipline: the VABS generator (lexicon, filter, selection, and guard code), the pre-registration documents with kill criteria for each mechanism variant, the generated vocabularies (including random-negative controls), and one JSON record per run (development-set grid, VOC-21 test arms for all four methods, COCO-Object and PASCAL-Context-60 transfer, per-class IoU for the safety criterion) are all included; the failed guard variants are released alongside the reported ones. During the RF-CLIP anchor runs, one class-name-file swap contaminated a concurrently launched official arm (exposed by the official and syn100 arms returning identical mIoU); the official file was restored from the upstream repository and both affected arms were re-run cleanly — all RF-CLIP numbers reported here come from the clean re-run, and the incident log is preserved in the release.

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